MANAS AGGRAWAL

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EDUCATION

Northeastern University **Boston, MA**
Masters of Science – Computer Science | GPA: 3.83 *Sep 2024 – Dec 2026*

- Graduate Teaching Assistant for Fundamentals of Software Engineering (CS 4530)
- Researching Multi-Agent LLM architectures, for e2e SDLC automation, across accuracy, cost efficiency

WORK EXPERIENCE

Hydrow **Boston, MA**
Software Engineer Intern *Jan 2026 – Present*

- Shipped the end-to-end backend for Hydrow's strength-training progress experience — 15+ REST APIs (NestJS/TypeScript/PostgreSQL) that let rowers track per-movement stats, personal records, and HydroMetrics fitness scores, all serving in under 25ms at production scale (6,500+ videos, 360+ movements, power users with thousands of workouts)
- Built 5+ badge checkers for Hydrow's gamification system with Redis-cached progress (1-hr TTL) so achievements unlock instantly — before background stats jobs complete — to drive workout engagement
- Profiled and eliminated two slow queries from the hot path (2,288ms and 1,646ms historical scans → effectively zero) by enriching a materialized view and checking only current-workout stats
- Replaced a synchronous JS gap-filling loop that blocked Node's event loop with a pure-SQL solution (generate_series + window functions), cutting progress-graph data prep to 0.724ms and keeping the API responsive under concurrent load
- Replaced offset pagination with Redis-backed cursor tokens across workout-history APIs, using canonical filter-key hashing to eliminate spurious cursor resets — delivering reliable infinite scroll and cutting redundant DB load
- Optimized workout-library filtering by rebuilding an instructor-compatibility query from 5 table scans to 1 (85ms) and adding accessory/difficulty filters across 6,500+ videos, so rowers only see workouts their equipment supports
- Delivered the full CRUD backend for interactive login-screen banners — multipart S3 upload, image validation, and transactional S3 cleanup on DB failure — powering promotional content shown to daily pre-login users

canAssist × Northeastern University **Boston, MA**
Forward Deployed Engineer & Scrum Master, Research Capstone *May 2026 – Present*

- Led a 5-person capstone team — drove client requirements, sprint ceremonies, and demos while building a NestJS/PostgreSQL/Redis backend instrumented with my own backend boilerplate and published telemetry package

Studio Graphene **Gurgaon, India**
Forward Deployed Engineer *Nov 2020 – Jul 2024*

- Built a serverless engineering-analytics platform on AWS Lambda to pinpoint bottlenecks (PR wait times, build failures, blocked dependencies), boosting engineering velocity by 23%
- Designed a Node.js/TypeScript REST backend serving 2M+ daily requests, with PostgreSQL partitioning/indexing to handle a 500GB+ catalog and real-time cross-region inventory sync
- Built event-driven microservices with AWS SQS for decoupled ingestion and processing, reducing response latency by ~40%

OPEN SOURCE PROJECTS

NestJs Backend Boilerplate | ***Nodejs, TypeScript, NestJs, Docker, Git, Zod, Biome, Winston***

- A production-ready boilerplate with all the essentials like JWT authentication, RBAC, validation, logger, database, Prisma ORM, Docker and Swagger OpenAPI Specs pre-configured. It has 51 stars and 11 forks so far

Telemetry NPM Package | ***Nodejs, TypeScript, AWS X-Ray, npm, OpenTelemetry, cloudwatch, Jaeger, Prometheus***

- Built and published a package on npm, enabling end-to-end distributed tracing across APIs, DB queries, and async tasks thereby reducing mean debugging time by ~60%
- It has over 1000 downloads on npm

TECHNICAL SKILLS

Languages:	JavaScript, TypeScript, Java, C++, Python
Frameworks:	Node.js, Express.js, NestJS, Django, Laravel, React
Database:	SQL, Postgres, MySQL, NoSQL, MongoDB, Elasticsearch, Amazon DynamoDB, Firebase
Cloud & DevOps:	Cloud Computing, AWS (Lambda, S3, SQS, DLQ, X-Ray), Docker, CI/CD, cloud infrastructure
Others:	Git, Sentry, OpenTelemetry, linux, REST APIs, web architecture, frontend, agile, SDLC